

Highlights

How much does the counting unit matter? Cross-country patent comparisons and the WIPO bulk file

Samiur Rahman Khan, Haseen Israa

- Office-summed WIPO publications are not invention counts.
- Office-sum inflation spans 1.05–4.48 across 68 economies (median pair 1.57).
- High D remains an EPC pattern after controlling for home-office share.
- Office ** tracks published PCT filings by origin (Spearman +0.991).
- Family scope shifts China–Singapore from about 500x to about 20x.

How much does the counting unit matter? Cross-country patent comparisons and the WIPO bulk file

Samiur Rahman Khan^{a,*}, Haseen Israq^b

^aIndependent Researcher, London, United Kingdom

^bBRAC University, Pragati Sarani, Dhaka, 1212, Bangladesh

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ABSTRACT

Open patent bulk files make cross-country comparisons accessible, but their rows do not necessarily represent inventions. This article measures how summing the World Intellectual Property Organization (WIPO) patent-publications-by-technology file across filing offices changes origin comparisons. For 68 economies with at least 500 publications in 2018–2022, and after separating the office code that behaves as the PCT international phase, the office-sum divided by the cell-wise leading office yields a duplication factor D from 1.05 to 4.48 (median pairwise ratio 1.57). Where the home office leads, $D = 1/H$ arithmetically; where it does not, D stays far below $1/H$ because foreign filings concentrate in a few large offices. The fraction of that identity realised, $D \cdot H$, has median 0.90 across 58 measurable-home economies. Partial Spearman of D with European Patent Convention (EPC) status controlling for H is +0.413: the high tail remains an EPC pattern after home-office share is held. Over seven non-EPC national-route origins the cell-wise leading office is closer to families than the office-sum (mean absolute relative error 0.101, or 0.069 excluding the year-anchor case, versus 0.803). That recommendation is not validated for the EPC structural case. Holding the family unit fixed, geographic scope still shifts the China–Singapore comparison from about 500 to about 20.

1. Introduction

Patent statistics often enter policy reports and cross-country research as if the word “patent” named a single, stable unit. It does not. An application, a publication, a granted right and a patent family are related records, but they answer different questions. This distinction has long been central to the measurement literature because patent data are indicators rather than direct observations of inventive activity [1, 2]. A publication count can describe the documentary output of an office. A first-filing or family count can approximate inventions. A count restricted to rights sought in several markets can instead indicate an internationally oriented subset. None is universally preferable; the appropriate unit follows from the construct being measured.

The problem becomes acute when an analyst works from an aggregate bulk file. Such files are attractive precisely because they remove many of the barriers to patent analysis: they are open, compact and already classified by country, year and technology. Their convenience, however, can hide the joins and counting rules that would be visible in document-level data. If a table has one cell for each applicant origin and filing office, summing its office column is easy. Interpreting that sum as a count of inventions is not. Related publications protecting one invention can be present

at several offices, so the sum measures publication-to-office assignments. The same procedure can affect origins differently because filing abroad is not equally common in every economy.

The WIPO IP Statistics Data Center, and the patent-publications-by-technology product in particular, is already used for cross-country work. A national intellectual-property office has produced a regional patent landscape from this file [3]. The same Data Center product appears in applied studies [4, 5, 6, 7]. Five uses, and none states how the office dimension is treated. That is a documented reporting gap, not an accusation of error. A companion regional analysis uses the family series for a different substantive purpose [8]; no findings from that study are required for the measurement presented here.

The established OECD and WIPO counting canon provides the conceptual remedy. Patent families group related filings; priority-based indicators move closer to the invention than office publication totals; and carefully selected international families can improve comparability when domestic institutional settings differ [2, 9, 10, 11]. International comparison nevertheless remains sensitive to attribution, field, year and geographic scope [12]. This literature explains why a unit must be chosen. It does not tell a user how large the origin-specific inflation is in the particular open bulk table already on their screen. Comparable source and classification choices have been shown to matter in applied patent studies, including work in *World Patent Information* that varies counting rules in PATSTAT

*Corresponding author

✉ samiurk70@gmail.com (S.R. Khan);
haseen.israq@g.bracu.ac.bd (H. Israq)
ORCID(s):

rather than in this bulk file [13]. Here, the source architecture itself is the object of study.

This article therefore asks a measurement question: how much does summing the WIPO patent-publications-by-technology bulk file across offices inflate an origin's count relative to its cell-wise leading office, and does that inflation behave like a common multiplier across origins? The issue is not whether every multi-office record can be deduplicated from the aggregate file; it cannot, because no family identifiers connect the office cells. The issue is whether a transparent diagnostic can reveal when an office-summed comparison is especially exposed to filing strategy.

The article makes four contributions. First, it defines what the office-sum in this WIPO file counts and, equally importantly, what it does not count. Second, it measures the duplication factor D against the arithmetic identity $D = 1/H$, and it reports that the high tail of D remains associated with membership of the European Patent Convention (EPC) after home-office share is controlled. Third, it shows that origin rankings move under equally defensible counting rules and, where family counts overlap, that the cell-wise leading office is the closer stand-in for families over the seven-economy overlap available here. Fourth, holding the family unit constant, it shows that geographic scope—all families versus foreign-oriented families—still changes origin comparisons. These contributions are respectively conceptual, diagnostic, practical and a matter of geographic scope.

The rest of the article follows the sequence required for a measurement problem. Section 2 defines the constructs and the bulk-file record. Section 3 describes the study design, sample and three quantities, including a worked China–Singapore comparison. Section 4 reports the descriptive, residual, rank and geographic-scope results. Section 5 discusses their use and limits for patent information work.

2. Counting units and the bulk file

2.1. Families, publications, and office-sums

A patent publication is a document made public by a patent authority at a particular procedural stage. It is an observable event in one jurisdiction, not a unique invention. The same underlying inventive subject matter may generate publications at the applicant's home office, at foreign offices, at a regional office and through the international phase of the Patent Cooperation Treaty (PCT). Publication statistics are consequently well suited to questions about office workload, documentary flows or routes of protection. They become problematic only when their unit is silently relabelled.

A patent family, by contrast, is a rule-based grouping of related patent documents. Family definitions vary because different grouping rules serve different analytical purposes [9]. A simple family may group

documents connected by a common priority; a broader construction can follow more complex priority links. Triadic families deliberately select inventions pursued at major offices and are designed for a different comparability problem [10]. The important point here is not that “family” is a perfect synonym for invention. It is that a family definition explicitly groups documents before counting, whereas an office-sum does not.

WIPO's family indicators used in this article are first-filing based and attributed by applicant origin [14]. Indicator 6a covers all observed families. Indicator 7 covers families extended to at least one foreign office. The latter is not a quality-adjusted version of the former and is not used here to declare one economy more inventive than another. It is a geographic-scope restriction that selects internationally extended protection. Such restrictions are common in patent-quality and international-comparison work because filing in additional markets reflects a selection decision as well as an opportunity to protect the invention [15, 16, 17].

An office-sum is a third object. For an applicant origin and period, it adds the publications recorded at each office. If one related set of documents appears at several offices, each office contributes to the sum. Without document or family links, the analyst cannot tell which cells relate to the same invention. The office-sum is therefore neither a simple family nor an international-family indicator. It is a measure of publication assignments over an office network.

2.2. The bulk file as an assignment table

The source examined here is WIPO's bulk indicator dataset of patent publications by publication year, filing office, applicant origin and technology field [18]. Each row represents one combination of those dimensions and supplies a count for that combination. The file is thus an aggregate assignment table, not a register in which each row is a patent. The analysed release contains 663,584 rows for 2000–2022 and 50.2 million publication-to-field assignments.

The assignment-table interpretation matters twice. Across offices, one invention can contribute publications in several jurisdictions. Across technology fields, one publication can contribute to several classified fields. The bulk value is therefore best read as a count attached to a specific origin–office–year–field cell. Collapsing one dimension does not automatically create unique documents along another. This is why a large, cleanly formatted total can still lack the construct implied by the label “inventions”.

Technology fields follow the Schmoch concordance used for country comparison [19]. The concordance allocates patent classifications to technology fields, and a document associated with several fields can contribute to each relevant field. The values in the analysed bulk file are integers throughout, consistent with whole counting in this particular table. The aggregate 50.2

million should consequently not be interpreted as the number of distinct patent documents. For the present study, the technology dimension is retained consistently in constructing every office quantity, so the concern is not a technology comparison. It is that the table contains assignments at more than one level.

This distinction also explains why family counting is not merely a final arithmetic step on the bulk file. Deduplication requires links. An aggregate cell reports how many assignments share its dimensions, but it does not report which assignment in one office corresponds to which assignment elsewhere. Document-level databases can construct families from priority relations. This bulk file can only support diagnostics that do not pretend those links are present.

2.3. Whole counting and the two ** tokens

Two identically printed tokens require separate treatment because they occur in different columns. In the applicant-origin column, ** denotes an unrecorded origin. Such rows cannot be assigned to an economy and are excluded from origin comparisons. In the filing-office column, ** behaves differently. It is smaller than the named-office sum within a slice and therefore does not behave as a grand-total field. Its volume is 309,728 in 2020, against WIPO's estimated 275,900 PCT applications filed in 2020 [20]. The two figures differ for two reasons: publications lag applications by about eighteen months, and whole-counted field assignments exceed document counts. The distribution of office ** is higher for origins that make heavier use of international filing.

WIPO's Data Center documentation states that international-phase PCT data are available under the PCT route, whereas subsequent national-phase activity is reported under patent or utility-model indicators [21]. In combination with the scale and origin pattern of office **, this supports treating that office token as the PCT international phase. Section 4.3 reports a discriminating origin-level rank check against published PCT applications by origin. The mapping is nevertheless inferential: the documentation available for the Data Center does not explicitly define the ** value in this bulk file's office column. This article therefore says that office ** *behaves as* the PCT layer. It does not claim that the token's meaning has been directly confirmed by a file-specific data dictionary.

Separating the token is necessary for a coherent office-sum. If it were added to all named offices, an international-phase publication would be treated as another office observation beside later national or regional publications. That may be appropriate when studying the use of the PCT route, but not when the stated outcome is publications at named patent offices. The origin token and office token are therefore removed or separated for different reasons: one lacks geographic

attribution; the other represents a distinct filing layer under the maintained interpretation.

Figure 1 states the counting problem as two illustrative filing routes. It is not an average family. The internationally filed route returns six, five, one, one and one under the five rules shown; the domestic-only route returns one, one, one, one and zero. Inflation is therefore a property of the route, not a common multiplier, and indicator 7 returns zero for a domestic invention rather than a smaller positive number.

3. Materials and methods

3.1. Study design and unit of analysis

The study is a descriptive measurement analysis of applicant-origin filing profiles. Its unit of comparison is an economy over the 2018–2022 window. The design does not estimate causal effects and does not rank national innovation systems. Instead, it asks whether the office distribution attached to one applicant origin makes an office-summed publication count more or less concentrated than the distribution attached to another origin.

The primary calculation begins with origin–year–technology-field cells. Named offices are compared within each cell, and the resulting counts are aggregated over the study window. Retaining a common cell structure prevents a country's technology composition or year mix from being silently collapsed at a different stage from another country's. Office ** is separated before the named-office duplication proxy is calculated. Rows with origin ** are excluded because they cannot enter an origin-specific result.

Three descriptive quantities are then placed in sequence. The home-office share describes how much of an origin's named-office activity is observed at its own office. The PCT-layer share describes the relative size of the separated ** office route. The duplication factor describes how dispersed the named-office count is relative to the cell-wise leading office. Together they distinguish domestic concentration, use of the inferred PCT route and multi-office dispersion. None directly observes family links.

3.2. Data sources and pin

Publication counts come from the WIPO patent-publications-by-technology bulk file, retrieved 30 July 2026 [18]. The cached input is pinned in the project manifest by SHA-256 so that the analysed release can be distinguished from later updates. The retrieval, transformation and figure scripts preserve the source file and generate the derived origin-level table. This pin is important for a live statistical service: a retrieval date alone identifies when access occurred, whereas the checksum identifies the exact bytes analysed.

Family counts come from a documented manual export of WIPO Data Center indicators 6a and 7,

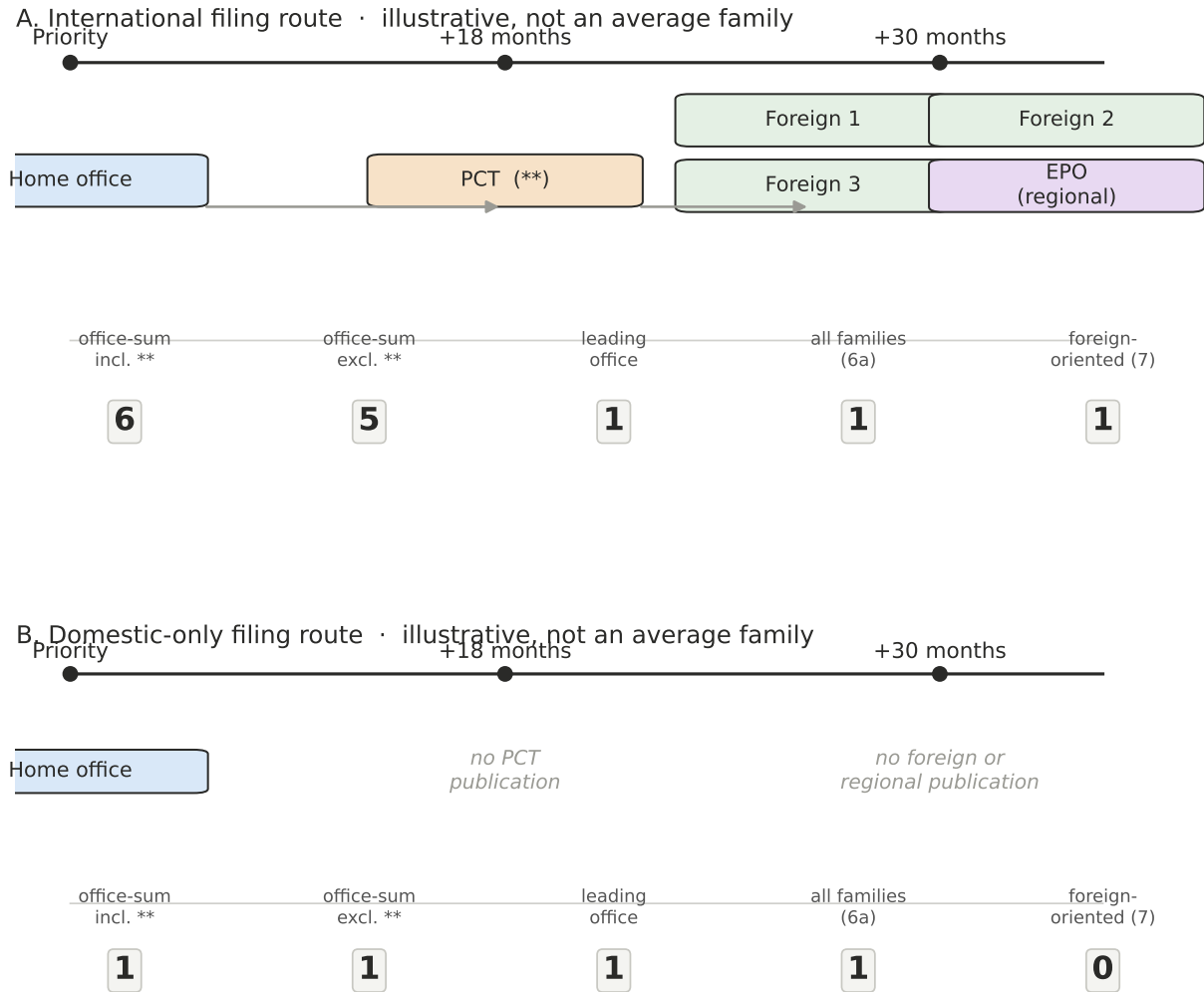


Figure 1: Two illustrative filing routes, not an average family size. Panel A follows a priority filing through a PCT international-phase publication, three foreign offices and a regional (EPO) publication. Panel B is a home-office publication only. The five counts under each panel are office-sum including **, office-sum excluding **, cell-wise leading office, all families (indicator 6a) and foreign-oriented families (indicator 7): 6/5/1/1/1 versus 1/1/1/1/0. The time axis marks priority, publication (eighteen months) and national-phase entry (thirty months). Source: authors' schematic.

total count by applicant's origin [14]. The Data Center recorded a database update in May 2026, and the export was retrieved 2 September 2026. The export route is manual because the family series is exposed through the Data Center interface rather than the publication bulk download. As a provenance check, the saved series reproduces, within source rounding, the China and Japan values printed for the same indicators in *World Intellectual Property Indicators 2024* [22].

The two data products are not merged record by record. The publication file is dated by publication year and the family series by earliest filing year. Their annual observations therefore have different temporal anchors, and one series is not assumed to nest inside the other. WIPO's Data Center documentation states that both the technology indicators and the family indicators are compiled from PATSTAT [21]. The two

products therefore share a lineage, so the non-nesting of annual observations is a dating-rule artefact rather than a difference of source. The family export is used first for a window-level consistency check and then for a separate geographic-scope illustration. These uses avoid presenting a same-year difference between the products as though it were a count of duplicates.

3.3. Sample construction

Sample construction begins with all 663,584 bulk rows. The analysis retains positive technology-field codes, excludes unrecorded applicant origins and forms origin-specific totals over 2018–2022. Economies enter the comparison when they have at least 500 publications in that window. The resulting sample contains 68 economies. The threshold screens volume, not attribution meaningfulness. Repeating the range calculation

with a threshold of 200 or 1,000 leaves the observed origin–year–field duplication-factor range at 1.05–4.48.

The home-office share requires a correspondence between applicant origin and an origin-resolved office row. This is measurable for 58 economies. For 10 economies, the bulk file has no origin-resolved row for the corresponding home office. That absence is not coded as a zero. A missing home-office block can reflect source coverage rather than no resident filing, so plotting it at zero would attach a behavioural interpretation that the data do not support. These economies remain in duplication and PCT calculations based on the observed offices, but they are excluded from the domestic-share distribution and from the horizontal axis of Fig. 2.

D is a property of the cell, not only of the file. The direction of coarsening is arithmetic: the sum of within-cell maxima is at least the maximum of the summed cells, so coarser cells cannot lower D . The magnitude is the finding. At origin–year–field granularity the range is 1.05 / 2.27 / 4.48. Collapsing the field dimension raises it to 1.05 / 2.51 / 5.41 at origin–year cells. Collapsing year while retaining field isolates the temporal-spread component: origin–field D is 1.05 / 2.39 / 4.90, so the maximum rises from 4.48 to 4.90 and the median from 2.27 to 2.39. Origin-only D is 1.05 / 2.54 / 5.77. That origin-only quantity is the office-sum divided by the largest single named office—the definition previously labelled and then corrected away from. Coarser cells therefore raise D ; they do not compress the 4.3-fold headline. This article reports the origin–year–field D as the main measure and the coarser range as a check. A missing home-office block can still raise the calculated proxy if the unobserved office would have been the cell-wise leader.

3.4. Three quantities and a worked example

Let p_{ioyt} denote the publication count for applicant origin i , named office o , publication year y and technology field t . Let $h(i)$ be the home office corresponding to origin i . The home-office share is

$$H_i = \frac{\sum_{y,t} p_{i,h(i),y,t}}{\sum_{o,y,t} p_{ioyt}},$$

where the denominator contains named offices only. The measure is undefined when the source lacks an origin-resolved block for $h(i)$. It is not replaced by zero.

Let q_{iyt} denote the count under office **. The PCT-layer share, under the inferential identification in Section 2.3, is

$$Q_i = \frac{\sum_{y,t} q_{iyt}}{\sum_{o,y,t} p_{ioyt} + \sum_{y,t} q_{iyt}}.$$

This denominator is deliberately different from that of the home-office share: Q_i asks how large the separate layer is when it is placed beside the named-office total.

Finally, define the duplication factor as

$$D_i = \frac{\sum_{y,t} \sum_o p_{ioyt}}{\sum_{y,t} \max_o(p_{ioyt})}.$$

The numerator is the named-office sum. The denominator adds the cell-wise leading-office count within each origin–year–technology–field cell. That denominator is the sum of within-cell maxima. It is therefore less than or equal to the origin-level maximum office, $\max_o(\sum_{y,t} p_{ioyt})$; the two denominators differ for 61 of 68 economies (median gap 0.224, maximum 2.294 at Monaco). The origin-only grain in Section 3.3 is exactly office-sum divided by that origin-level maximum, which is why the denominators differ. Computed D is the lower of the two ratios. The factor is bounded below by unity and rises as meaningful volume is spread across additional offices. A regional office, including the European Patent Office, enters as one observed office; no equivalent-count multiplier is applied.

The duplication factor confounds office dispersion with the temporal spread of publication, because related publications that land in different years occupy separate cells and therefore add separate cell-wise maxima. The origin–field grain, collapsing year while retaining field, isolates that timing component: the maximum rises from 4.48 to 4.90 and the median from 2.27 to 2.39.

When the home office is the cell-wise leader, $D = 1/H$ exactly, up to cells where another office leads. China, Korea and Japan sit on that identity ($H = 95.4\%$, 68.1% and 55.4% ; $D = 1.05$, 1.47 and 1.80). Singapore's home office is not the leader—the United States office is—so the identity does not bind: $H = 5.0\%$, $1/H = 20.08$, $D = 2.60$ and residual 17.48.

China provides the domestic-concentration case. In 2018–2022, 95.4% of its named-office publications are at its home office, and its duplication factor is 1.05. The worked reading is straightforward: within the bulk file, the named-office sum is only modestly larger than the contribution of the cell-wise leading office. This does not prove that the leading-office count is a family count. It shows that adding other observed offices makes little difference to this particular office-based total.

Singapore provides the contrasting filing profile. Its home-office share is 5.0%, so its own office cannot stand in for most publications attributed to Singaporean applicants in the file. Its activity is distributed across foreign offices, and the office-sum accumulates those observations. The contrast with China is therefore not an arithmetic nuisance that a common percentage adjustment can remove. The two origins place their publications through different office networks.

The sample maximum further clarifies the denominator. Switzerland has a duplication factor of 4.48, a home-office share of 3.1%, and its origin-level leading office is the EPO rather than its home office. The factor does not say that each Swiss invention has 4.48

publications, nor does it place an exact bound on distinct inventions. It says that, under the cell-wise construction, the named-office sum is much larger relative to the leading-office baseline than it is for China. The measure is consequently labelled a duplication *proxy*.

3.5. Check against WIPO family indicators

A useful proxy should at least order filing profiles similarly to a measure that uses families, even though the levels need not coincide. The check compares D_i with publications divided by WIPO indicator 6a families over the common 2018–2022 window. Seven economies overlap: China, Japan, Korea, Malaysia, the Philippines, Singapore and Thailand—all non-EPC origins that file through national rather than regional-first routes. Spearman correlation is used because the diagnostic claim concerns ordering rather than a linear conversion.

The two ratios are not independent measures. D is the office-sum divided by the cell-wise leading office; publications per family uses the same office-sum as its numerator. Rank agreement is therefore expected to be high if the mechanism exists at all. The practical comparison is whether the leading-office count or the office-sum lies closer to families. Publications and families use different dating rules, and D cannot fall below unity whereas the observed publications-to-families ratio can. The exercise is a consistency check on behaviour, not an estimated deduplication formula. Family exports for four EPC origins were not available in the pinned input. Section 4.5 reports the coefficients and the error comparison.

3.6. Geographic-scope comparison: indicators 6a and 7

The final analysis holds the family unit constant and changes geographic scope. WIPO indicator 6a counts all families by applicant origin and earliest filing year. Indicator 7 counts the subset extended to at least one foreign office. China and Singapore are used because their domestic shares in the bulk file already establish sharply contrasting filing profiles. Their family series from 2000–2022 show whether the substantive comparison also depends on including or excluding domestic-only families.

The all-family and foreign-oriented ratios are not alternatives from which the article selects a preferred headline. They answer different questions. The first describes the full first-filing-based family portfolio observed for an origin. The second describes the internationally extended subset. Placing them together demonstrates that correcting the unit does not eliminate the need to define scope.

4. Results

4.1. Sample and range table

Table 1 summarises the three bulk-file quantities. Across the 58 economies whose home office is measurable, the home-office share ranges from 0.3% (Saudi Arabia) to 95.4% (China), with a median of 37.8% (p25 9.2, p75 64.8). Across all 68 economies, the inferred PCT-layer share ranges from 3.2% (Macao SAR) to 54.4% (Iran) and has a median of 15.1% (p25 12.7, p75 18.6). Both PCT endpoints are origins whose home office is unmeasurable. The duplication factor ranges from 1.05 to 4.48. Its all-economy median is 2.27 (p25 1.55, p75 3.34); among the 58 economies with a measurable home office, it is 2.14 (p25 1.48, p75 3.45).

The duplication proxy does not apply a common uplift across origins. The more robust summary is the median pairwise ratio of 1.57, the median of the 2,278 pairwise ratios D_i/D_j taken larger over smaller. The 4.3-fold figure is the ratio of the sample ends, Switzerland versus China. These are differences in D_i , not estimates of the number of copies attached to an invention. They establish the empirical range over which the same office-summing operation behaves differently in this file. The pooled median of 2.27 is a composition artefact: 35 of the 68 are EPC contracting-state origins. Group medians on the 58 economies with a measurable home office are the more stable summary.

4.2. Residual from the identity $D = 1/H$

Figure 2 places the duplication factor against the home-office share for the 58 measurable economies, with the curve $D = 1/H$ overlaid. Nineteen of those 58 lie within 0.05 of the identity. Thirty-five of 58 have a home office that is the origin-level leading named office. All 19 are among those 35. The remaining 16 lead at origin level but not in every year-field cell, so cell-wise D sits below $1/H$. China occupies the low-duplication, domestic-heavy end at 95.4% and 1.05, where the identity binds. The information in the scatter is not a downward-sloping cloud. It is the gap below $1/H$. At low home-office share the identity predicts D far above the observed ceiling.

The residual $1/H - D$ inherits that explosion: its maximum is set by how small the smallest H happens to be. Across the 58 it runs from 0.00 to 369.68. The scale-free counterpart is $D \cdot H$, the fraction of the identity actually realised, bounded in $(0, 1]$. It has median 0.90 (p25 0.32, p75 0.99); 23 of 58 realise at least 0.95 of it. Economies on the identity realise essentially all of it (China, Korea and Japan at 1.00; Malaysia at 0.96). The low- H group does not: Singapore 0.13, Switzerland 0.14, Saudi Arabia 0.005. Economies that file away from home do not disperse across all offices. They concentrate in a handful of large ones, so office-sum inflation is bounded far below what home-office share alone would imply.

Table 1

Filing-profile measures in the WIPO bulk file, 2018–2022. The sample contains 68 economies with at least 500 publications; the home-office row contains the 58 economies for which the own office is origin-resolved. The fourth row reports D on that measurable-home base. The fifth row is the fraction of the identity $D = 1/H$ actually realised. Per-economy values are in `counting_generality.csv` in the replication archive. Source: authors' calculations from the WIPO bulk patent-indicators file.

Measure	n	Min.	p25	Median	p75	Max.
Home-office share (%)	58	0.3	9.2	37.8	64.8	95.4
PCT-layer share (%)	68	3.2	12.7	15.1	18.6	54.4
Duplication factor	68	1.05	1.55	2.27	3.34	4.48
Duplication factor (measurable home)	58	1.05	1.48	2.14	3.45	4.48
$D \cdot H$ (measurable home)	58	0.005	0.32	0.90	0.99	1.00

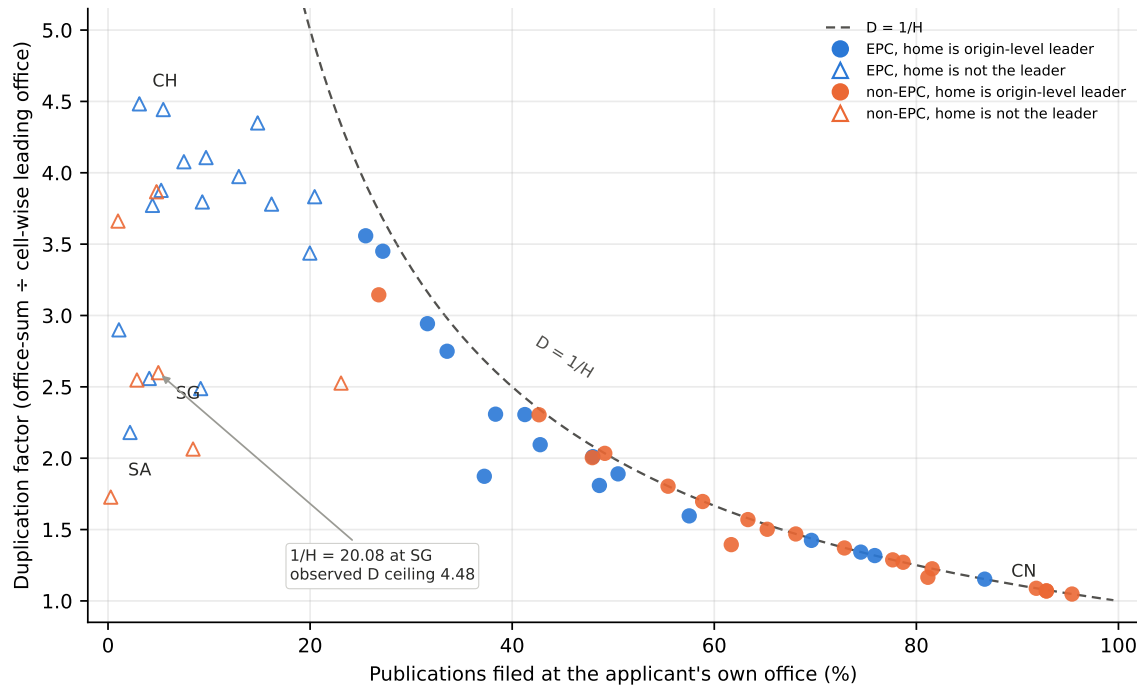


Figure 2: Duplication factor versus the applicant-origin home-office share, 2018–2022, for the 58 economies with a measurable home office. The dashed curve is the identity $D = 1/H$, which is arithmetic when the home office is the cell-wise leader. Marker fill denotes whether the home office is the origin-level leading named office; colour denotes EPC versus non-EPC origin. At low H , Singapore has $1/H = 20.08$ and $D = 2.60$; Saudi Arabia has $1/H = 371.41$ and $D = 1.73$; the observed ceiling is 4.48. Ten economies remain in the range statistics but are omitted here because the home office is not origin-resolved. Source: authors' calculations from the WIPO bulk patent-indicators file.

Saudi Arabia has $H = 0.3\%$, $1/H = 371.41$, $D = 1.73$ and residual 369.68, with 32 origin-resolved home-office publications. That count places it at the boundary of the measurable/unmeasurable distinction rather than clearly inside it: a partially resolved home-office block also produces a small non-zero count, and 32 publications over five years cannot carry the paper's largest single number as a settled filing pattern. Among economies with at least 1,000 origin-resolved home-office publications, the largest residual is Israel: $H = 2.9\%$, $1/H = 34.95$, $D = 2.55$ and residual 32.40, on 1,450 home publications. Singapore remains the interpretable mid-case: $1/H = 20.08$, $D = 2.60$ and residual 17.48.

The high tail of D remains an EPC pattern after home-office share is held. Partial Spearman of D with EPC status controlling for H is $+0.413$ ($p = 0.0016$; 95% CI $[0.171, 0.608]$) across the 58. Table 2 shows where that association sits: in the lowest H quartile, median D is 2.57 among six non-EPC origins and 3.77 among nine EPC origins. The gap narrows in the upper half of H . The second quartile's non-EPC cell has only two origins and is not a basis for a separate claim. At the lowest H the values still interleave—New Zealand 3.87 beside Sweden 3.88—and none of those four matched- H origins is led by the EPO (all four are led by the United States office). EP-as-leader is therefore not required for high D . The matched

Table 2

Median duplication factor by home-office-share quartile and EPC status, 2018–2022, measurable-home base ($n = 58$). Quartiles are of H . The second-quartile non-EPC cell has $n = 2$. Source: authors' calculations from the WIPO bulk patent-indicators file.

Quartile	n	non-EPC D	n	EPC D
Q1 (low H)	6	2.57	9	3.77
Q2	2	2.84	12	3.67
Q3	7	1.80	7	2.01
Q4 (high H)	11	1.23	4	1.33

Table 3

Duplication factor by EPC status, 2018–2022. The measurable-home base of 58 is the primary comparison; the all-68 rows include ten origins whose home office is not origin-resolved. Source: authors' calculations from the WIPO bulk patent-indicators file.

Base	Group	n	Min.	Median	Max.
58	EPC	32	1.15	2.82	4.48
58	non-EPC	26	1.05	1.63	3.87
68	EPC	35	1.15	2.75	4.48
68	non-EPC	33	1.05	1.73	4.10

comparison does not replace the partial correlation; it shows that the residual EPC association is a shift in the low- H distribution, not a requirement that the regional office lead.

Table 3 records the group ranges. On the measurable-home base the non-EPC maximum is New Zealand. The five highest values on that base are all EPC: Switzerland, Luxembourg, Norway, Belgium and the Netherlands. Ten of the 68 origins have the EPO as origin-level leading office; all ten are EPC origins. The all-68 non-EPC maximum is Antigua and Barbuda, whose home office is unmeasurable.

4.3. The PCT layer

The separated ** office layer also varies by applicant origin. Its sample range is 3.2% (Macao SAR) to 54.4% (Iran), with a median of 15.1%. Iran routing over half of its publications through office ** is the sample maximum; because its home office is unmeasurable, that endpoint is taken up with Antigua and Monaco in Section 5.3 rather than treated here as a confirmed filing behaviour. In the China–Singapore comparison, the layer accounts for 3.9% and 20.2%, respectively, over 2018–2022. Adding this layer to named-office publications would therefore change the two origin totals by different proportions.

If office ** is the PCT international phase, the PCT-layer share Q should rise as the home-office share H falls. Over the 58 measurable economies, Spearman correlation of Q with H is -0.559 ($p = 0.000005$; 95% CI $[-0.714, -0.352]$). The relationship is descriptive,

Table 4

PCT-layer share of publications for selected origins in the 68, 2018–2022, $\text{tec_id} > 0$, under the same 68-economy filters as the range table; not the all-year audit table. Office ** is treated as the PCT international phase on behavioural grounds and on the origin-level rank check in Section 4.3; the mapping is still without a file-specific dictionary. Source: authors' calculations from the WIPO bulk patent-indicators file.

Origin	PCT-layer share (%)
Russia	4.5
China	3.9
Korea	8.8
India	7.5
Japan	14.3
Germany	13.7
United States	13.3
Netherlands	16.6
Sweden	18.8

not confirmation: H and Q share the named-office sum in different roles in their denominators, and the scatter is fat. It also does not discriminate against the rival reading. If ** were an unresolved-office bucket, unresolved-office publications would still come disproportionately from origins whose filings are dispersed, which is the same low- H group. Both readings imply a negative Q – H correlation.

A rank check against published PCT applications by origin does discriminate. If office ** is the international phase, origin-level ** volumes should track WIPO's published PCT filings; an unknown-office bucket would have no reason to. Spearman correlation of 2020 office-** publications against published 2020 PCT applications by origin is $+0.991$ over the twenty largest named origins ($p < 10^{-15}$; 95% CI $[0.977, 0.997]$), using the same 2020 office-** construction as the 309,728 aggregate [20]. Sweden matches exactly: 4,356 publications under ** and 4,356 published PCT applications. The Q – H correlation is consistent with the PCT identification without ruling out the alternative; this origin-level check is the test that would have failed if ** were merely unresolved offices.

Table 4 reports the PCT-layer share for selected origins in the 68, after that correlation. The values use the same $\text{tec_id} > 0$, 2018–2022 and 68-economy filters as the rest of the paper; they are not the all-year audit table. Russia (4.5%) and China (3.9%) sit below Korea (8.8%) and India (7.5%), which in turn sit below Japan (14.3%), Germany (13.7%) and the United States (13.3%); the Netherlands (16.6%) and Sweden (18.8%) are higher still.

This result is distinct from Fig. 2. The duplication factor describes dispersion across named offices after the token has been separated. The PCT-layer share describes the size of the removed route relative to the

combined total. An analyst interested in PCT use may retain and report it as an outcome. An analyst describing national or regional office publications should separate it and state that treatment. Neither choice converts publications into families.

4.4. Rank instability

Figure 3 ranks the 25 largest origins by named-office publication volume under three rules, each legitimate for a different question: the office-sum including **, the office-sum excluding **, and the cell-wise leading office. None is the correct ranking. The figure is restricted to the top 25 by named-office volume because the full 68 is unreadable, not in order to hide a larger movement outside that window. Among those 25, Russia ranks 13th on the office-sum including ** and 6th on the cell-wise leading office, because its PCT-layer share is 4.5% and its domestic home-office share is 91.9%, so removing ** and moving to the leading office penalises internationally dispersed origins more than Russia. Türkiye ranks 23rd versus 15th; Brazil 24th versus 16th; Poland 26th versus 18th. The largest movement among the 68 is Antigua and Barbuda, 50th versus 65th, a first-named-applicant domicile case taken up in Section 5.3. The figure is a consequence of the range in Table 1: if D is not a common multiplier, origin order is not invariant to the counting rule.

4.5. Proxy versus families

Figure 4 places WIPO all-family counts on a logarithmic horizontal axis against the cell-wise leading office (circles) and the office-sum (squares) for the seven overlapping economies. The dashed line is 1:1. The leading-office series tracks that line; the office-sum series lies above it, except where dating rules pull a point toward or below equality. The Philippines office-sum can sit near or below the 1:1 line because publication year and earliest filing year do not nest; that is a feature of the dating rules in Section 3.2, not an anomaly in the Philippines series.

Spearman +1.000 between the cell-wise leading office and families on $n = 7$ is a size ranking, not an independent test. The office-sum against families returns Spearman +0.964 ($p = 0.0005$; 95% CI [0.771, 0.995]), the same coefficient as D against publications/families, because those ratios share the office-sum numerator. Rank agreement barely discriminates the two candidates. What separates them is the mean absolute relative error: 0.101 for the leading office versus 0.803 for the office-sum, or 0.069 versus 0.803 once the Philippines is excluded. The year-anchor offset—publication year versus earliest filing year—applies to all seven economies. It is visible in the Philippines because that origin's high home-office share puts its ratio closest to unity to begin with (leading-office/families 0.70). Singapore has $|\text{lead/fam} - 1| = 0.22$, the largest remaining error. Spearman correlation of (leading-office/families) with D is -0.214 ($p = 0.6445$): there is no evidence at

$n = 7$ that D predicts the error a user would incur, and D is not a validated warning light for that proxy error. Seven points support a scoped recommendation for this overlap, all non-EPC origins filing through national rather than regional-first routes. They do not support a general calibration equation, they do not validate the recommendation for the EPC structural case, and they do not support dividing an office-sum by D_i and calling the result families.

4.6. China–Singapore on all families versus foreign-oriented families

Figure 5 reports the China-to-Singapore ratio while holding the family unit constant. China's all-family series rises from 23,079 in 2000 to 1,517,232 in 2022; Singapore's rises from 716 to 3,036. On all families, the 2022 ratio is about five hundred. When the scope is restricted to families extended abroad, the ratio is about twenty. The vertical axis is logarithmic so that the foreign-oriented series remains readable. The 2019 dip in the all-family series is in the data (ratio 416). Values for 2022 are provisional because later foreign members may not yet be observed.

The difference corresponds to the composition of the two portfolios. In 2022, 3.6% of Chinese families are observed as extended abroad, compared with 91.9% of Singaporean families. Because recent first filings may acquire foreign members after their earliest filing year, the 2022 foreign-oriented result is provisional. The more mature 2018 anchor gives the same qualitative contrast: about 470 on all families and about 15 on foreign-oriented families.

5. Discussion

5.1. What an office-sum measures

Three findings do not depend on the scoped checks that follow. First, the office-sum is not an invention count. Relative to the cell-wise leading office, inflation ranges from 1.05 to 4.48 across 68 economies, with a median pairwise ratio of 1.57; it is not a common multiplier. Second, where the home office leads, D is arithmetically $1/H$, so most observed variation in D is a restatement of home-office share. Where it does not lead, D stays far below $1/H$ because foreign filings concentrate in a few large offices, and $D \cdot H$ is the fraction of that identity the data deliver. Third, origin rankings are not invariant to defensible counting rules, demonstrated on the top 25. Everything else in this section is scoped.

The office-sum is not meaningless; it is misnamed when presented as inventions. Properly labelled, it describes the volume of publication assignments generated by applicants from an origin across the offices represented in the bulk file. That volume combines at least two components: the scale of underlying patenting and the breadth of the office network through which protection is sought. For questions about documentary

How much does the counting unit matter?

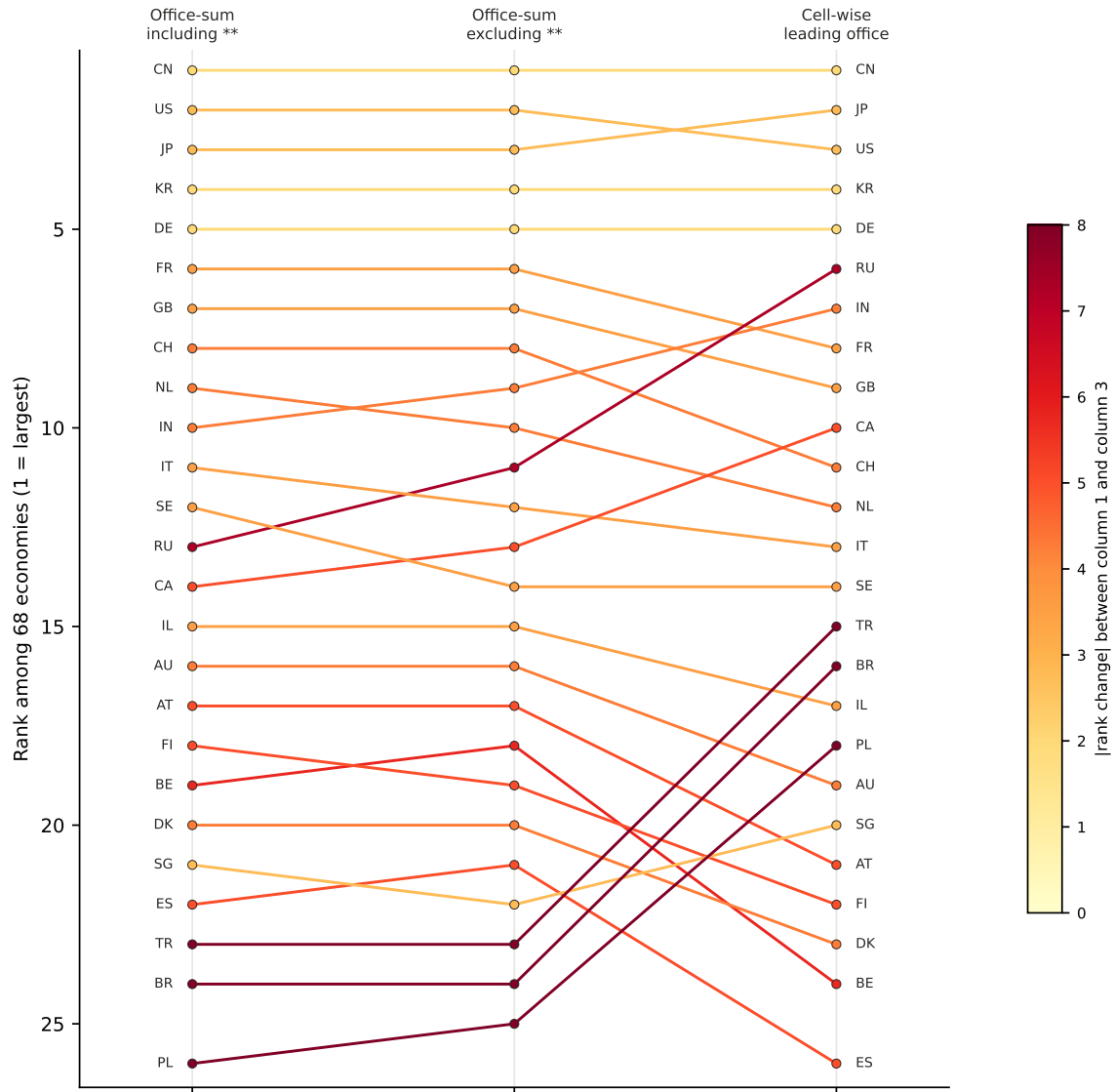


Figure 3: Rank of the 25 largest origins by named-office publication volume under three counting rules, 2018–2022. The columns are the office-sum including **, the office-sum excluding **, and the cell-wise leading office. All three are legitimate answers to different questions; none is the correct ranking. The colour bar reaches 8. Among this top 25, Russia ranks 13th (sum including **) versus 6th (cell-wise leading office); Türkiye 23rd versus 15th; Brazil 24th versus 16th; Poland 26th versus 18th. The figure is restricted to the top 25 because the full 68 is unreadable. Antigua and Barbuda, 50th versus 65th, is the largest movement among the 68. Source: authors' calculations from the WIPO bulk patent-indicators file.

reach, filing routes or the international workload associated with an origin, the combination may be the object of interest. For comparisons of inventive output, it is a confounder.

The empirical range shows why this is more than a terminological preference. A common multiplier would preserve origin ratios even while making every level too large. The measured proxy is not common. The median pair differs by 1.57; China and Switzerland sit at 1.05 and 4.48, a 4.3-fold difference at the sample ends. Office summing therefore embeds an origin-specific filing profile. Much of that profile is arithmetic: where the home office is the cell-wise leader, $D = 1/H$.

The remainder is structure. Observed D lies below $1/H$ because foreign filings concentrate in a few large offices. Partial Spearman of +0.413 after controlling for H is the evidence that the high tail remains an EPC pattern rather than a generic international-orientation gradient; the matched- H interleave (New Zealand beside Sweden) shows that EP-as-leader is not required for that tail. The pooled median of 2.27 is a mixture: 35 of the 68 are EPC contracting-state origins, and the group medians on the measurable-home 58 are the more stable summary. The direction of cell coarsening is arithmetic—coarser cells cannot lower D —and the magnitude is the finding: origin-only

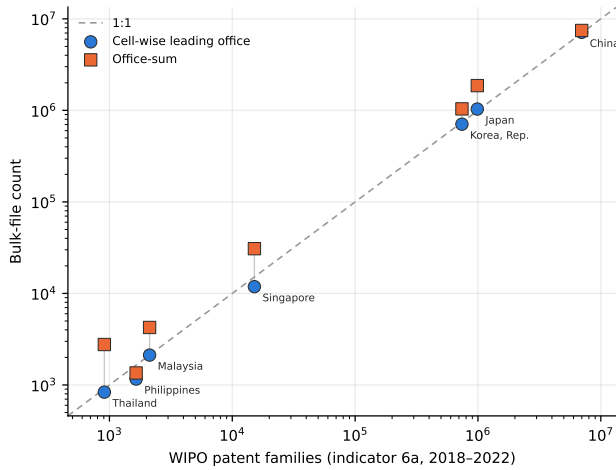


Figure 4: WIPO patent families (indicator 6a) on a logarithmic horizontal axis against the cell-wise leading office (circles) and the office-sum (squares), 2018–2022, for the seven economies available in both sources. The dashed line is 1:1. A faint vertical connector joins each pair; that gap is that economy's D . The Philippines office-sum can sit near or below the 1:1 line because publication year and earliest filing year do not nest. Source: authors' calculations from the WIPO bulk patent-indicators file and WIPO indicator 6a.

D reaches 1.05 / 2.54 / 5.77, which is the office-sum divided by the largest single named office.

This does not imply that the cell-wise leading office supplies the true invention count. China's factor of 1.05 says that other named offices add little relative to the leading-office baseline in these cells. It does not say that the Chinese home-office total contains one record per invention. The baseline can include multiple procedural publications, whole-counted technology assignments and other features of the source. Likewise, Switzerland's 4.48 is not an exact average family size. The diagnostic compares concentration under one consistent aggregation rule; it does not recover hidden record linkage.

The PCT result adds a separate source of non-comparability. Office ** tracks published PCT applications by origin, but its share is not constant across origins. In the worked pair, Singapore's 20.2% is far above China's 3.9%. A total formed by adding that route to all named offices would therefore alter the comparison again. The correct treatment depends on the question, but the choice cannot remain implicit.

The family analysis then establishes a second distinction. Moving from office publications to patent families addresses the multi-office assignment problem because related documents are grouped under a declared family rule. It does not decide whether domestic-only families belong in the construct. The China–Singapore ratio in 2022 is about five hundred for all families and about twenty for the foreign-oriented subset. Neither ratio invalidates the other. They describe different geographic scopes, and their distance reflects

the contrast between 3.6% and 91.9% foreign extension in the two portfolios.

5.2. Using the file

For a patent-information practitioner, the safest workflow begins before aggregation, with a sentence that names the intended construct. If the question concerns publications handled by a specified office, the office dimension should be fixed and the resulting statistic labelled as publications. If the question concerns an origin's use of the international system, a multi-office sum or the PCT layer may be informative, provided the unit remains publication assignments and the routes are reported separately. If the question concerns inventions, the analysis needs a family product or document-level data from which an explicit family definition can be built.

This sequence is especially important in reusable dashboards and country rankings. A code path that groups by applicant origin and sums every remaining numeric column can be technically correct yet conceptually wrong. The error will not be visible as a missing value or an implausible total. Fig. 3 shows the consequence: equally defensible rules reorder the same origins. Metadata and method notes therefore need to travel with the exported table, not remain in the analyst's private understanding.

When only this file is available, the cell-wise leading office is closer to families than the office-sum over the seven-economy overlap, all non-EPC origins filing through national rather than regional-first routes. The validation excludes the EPC structural case. Singapore is the only low- H analogue in that overlap and has the largest genuine error. Four EPC family exports were not available. The recommendation is therefore scoped to this overlap; it is not stated for regional-first filing systems. D predicts the office-sum's distance from the leading office by construction. There is no evidence at $n = 7$ that it predicts the error a user would incur, so it is not a validated warning light for that proxy error. It does not support dividing an office-sum by D_i and calling the result families.

Users should also preserve the distinction between the two ** tokens in processing code and documentation. Dropping both with one unexplained filter would conceal why observations were removed. Origin ** cannot support country attribution. Office **, under the interpretation used here, is analytically meaningful as the PCT layer. A reproducible transformation should name the column and reason for each treatment.

Finally, a family total should carry its scope in the table heading. "Patent families by applicant origin" is incomplete if the reader cannot tell whether domestic-only families are included. Indicator 6a is suitable when the full origin portfolio is intended. Indicator 7 is suitable when the construct is internationally extended

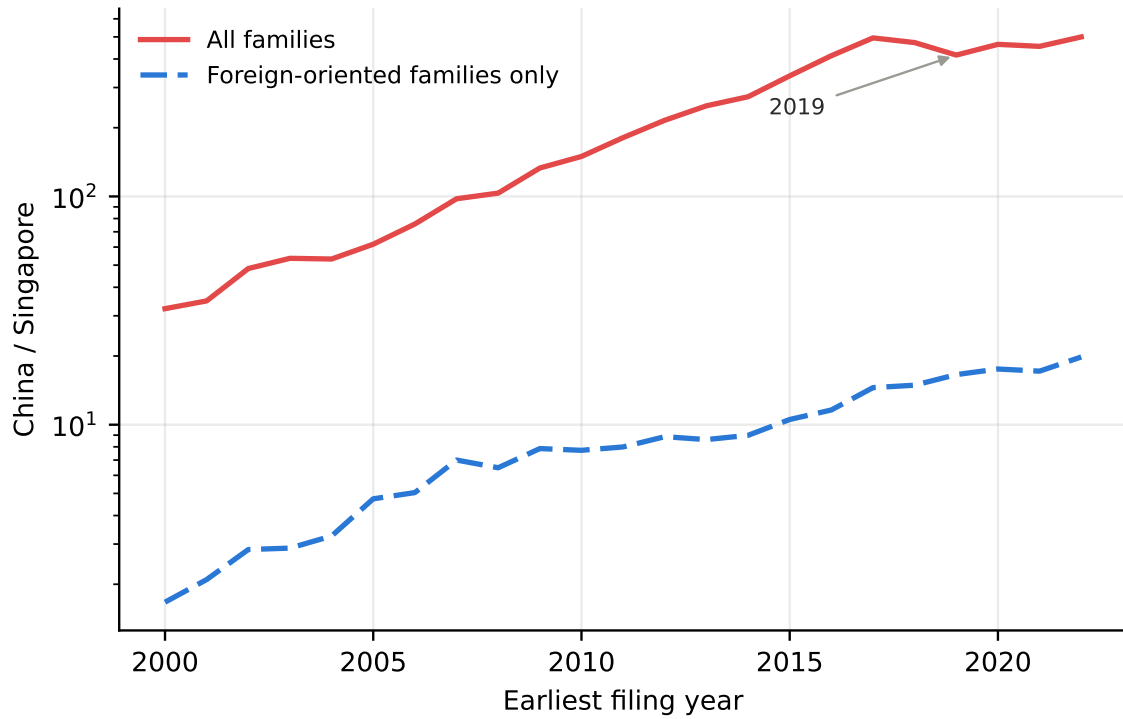


Figure 5: China-to-Singapore patent-family ratio under two geographic scopes, by earliest filing year, 2000–2022, on a logarithmic vertical axis. The solid series uses all families (WIPO indicator 6a); the dashed series uses only families extended to at least one foreign office (indicator 7). The 2019 dip in the all-family series is in the data (ratio 416). Values for 2022 are provisional because later foreign members may not yet be observed. Source: authors' calculations from WIPO IP Statistics Data Center indicators 6a and 7.

inventions. International families, including triadic constructions, can improve comparability for some research questions but also select on filing strategy and market reach [10, 17]. The scope should follow the question rather than being treated as a generic quality filter.

5.3. Limitations

The first limitation concerns identification of the PCT layer. Origin-level office-** volumes track published PCT applications by origin (Spearman +0.991 over the twenty largest named origins in 2020), which is the test an unknown-office bucket would fail. A file-specific dictionary that explicitly maps this token has still not been located. All PCT-layer findings are therefore stronger than a $Q-H$ correlation but still without a dictionary confirmation. A different documented meaning would require the layer to be renamed and reinterpreted.

Second, the duplication factor is intentionally modest in what it claims. The aggregate bulk file lacks identifiers linking related publications across offices, so D_i cannot distinguish an international family from unrelated documents that happen to share a cell. It cannot identify continuations, procedural publications or family structures. Its denominator is a cell-wise leading-office baseline, not a proven count of inventions, and 4.3 is a fold-difference between the proxy's sample

ends rather than a deduplication bound. D also confounds office dispersion with the temporal spread of publication; the origin-field grain bounds that timing component.

Third, the comparison with family indicators contains seven economies, all non-EPC origins filing through national rather than regional-first routes. The leading-office comparison discloses a scoped practical recommendation, but the exercise is not a validation study across the full sample and it excludes the EPC structural case the paper itself identifies. Family exports for four EPC origins were not available. In particular, the rank check does not establish a conversion from the proxy to family counts, and D is not a validated warning light for the error a user would incur. More overlapping origins or record-level family links would be needed to test general calibration.

Fourth, whole counting presents a documentation tension. The bulk-file values are integers and the Schmoch concordance supports assignment to multiple technology fields, while WIPO's general Data Center description refers to fractional counting for technology indicators [19, 21]. This article reports the file as observed and retains a consistent field-cell construction. It cannot resolve whether the general documentation was intended to describe this exact download. This

uncertainty reinforces the need to call the 50.2 million total assignments rather than documents. Whole-versus-fractional counting and the identification of office ** are both questions about how one PATSTAT compilation was run [21].

Fifth, WIPO attributes origin to the residence of the first-named applicant and ignores co-applicants [21]. Multi-jurisdiction filings by multinational groups can therefore carry different first-named applicants at different offices, so the same invention may be attributed to different origins in different office cells. That attribution rule places micro-state and offshore-domicile origins at the extremes of the structural measures this file produces. Antigua and Barbuda has $D = 4.10$, origin-only $D = 5.77$, 1,592 named-office publications and Japan as origin-level leading office; its home office is unmeasurable, and it is the all-68 non-EPC maximum and the largest rank movement among the 68 (50th versus 65th). Monaco has the largest labelled-versus-computed D gap, 2,294, with $D = 2.56$, the EPO as origin-level leader, a measurable home office and EPC membership. Iran has the sample-maximum PCT-layer share, 54.4%, and an unmeasurable home office; it belongs in this class rather than in the PCT-layer result as a confirmed filing behaviour (Section 4.3). None of these is an inventing-economy profile in the ordinary sense; they follow from first-named-applicant attribution and from unmeasurable home offices. The 500-publication threshold does not screen this, because it screens volume rather than attribution meaningfulness. Singapore remains a hub case whose publications are distributed across foreign offices, but it is no longer the obvious illustration of the attribution rule.

Sixth, the family tail is subject to right truncation. Families are dated by earliest filing year, but foreign members can appear later. The 2022 foreign-oriented shares and ratio are therefore provisional, and the 2018 anchor is reported as a more mature comparison. This issue does not affect the 2018–2022 publication-window range in the same way because those observations are dated by publication year, but it limits direct annual alignment between products.

Finally, the home office is unmeasurable for 10 of the 68 economies. Retaining their observed-office duplication factors preserves sample coverage, but a missing home-office block may omit the true cell-wise leader and raise the proxy. Saudi Arabia's 32 origin-resolved home-office publications place it at that boundary rather than clearly inside the measurable set (Section 4.2). The separate median for the 58 measurable economies and the exclusion of missing shares from Fig. 2 make that limitation visible. Neither the threshold sensitivity, the cell-grain check nor the reported medians can replace origin-resolved source coverage.

6. Conclusions

Three findings hold without the scoped qualifications that follow. The office-sum is not an invention count: relative to the cell-wise leading office, D ranges from 1.05 to 4.48 across 68 economies (median pairwise ratio 1.57), and it is not a common multiplier. Where the home office leads, that range is the identity $D = 1/H$; where it does not, D stays far below $1/H$ because foreign filings concentrate in a few large offices. Origin rankings are not invariant to defensible counting rules, demonstrated on the top 25.

The WIPO patent-publications-by-technology bulk file supports useful cross-country work when that assignment structure is respected. Partial Spearman of D with EPC status controlling for H is +0.413, so the high tail remains an EPC pattern after home-office share is held; EP-as-leader is not required for it. Office ** tracks published PCT applications by origin (Spearman +0.991) and should be reported according to the question being asked. Over the seven-economy overlap—all non-EPC origins filing through national rather than regional-first routes—the cell-wise leading office is closer to families than the office-sum (mean absolute relative error 0.101, or 0.069 excluding the year-anchor case, versus 0.803). That recommendation is not validated for the EPC structural case; Singapore is the only low- H analogue in the overlap and has the largest remaining error (0.22). Once families are used, geographic scope must still be defined: all families and foreign-oriented families produce different but legitimate China–Singapore comparisons (about 500 versus about 20). Interpretable patent statistics therefore require three explicit choices—counting unit, treatment of the PCT layer and family scope—before the country totals are compared.

Data availability

The bulk file and the family export are public. The derived tables, figure files and machine-readable record of numeric claims are archived with the analysis pipeline at <https://github.com/samiurk70/ASEAN-vs-ASIA-Patent-Analysis>. Per-economy home-office share, PCT-layer share and duplication factor for the 68 economies are in `counting_generality.csv` in that archive. A Zenodo snapshot will be linked here when minted.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI-assisted editor in order to organise the draft from the authors' analysis pipeline and to match the Elsevier CAS template. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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CRedit authorship contribution statement

Samiur Rahman Khan: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft. **Haseen Israq:** Validation, Writing – review and editing.

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